Chapter 16 — Lab Safety and Practical Skills

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Laboratory safety, hazards and waste disposal

Q.1 Describe general laboratory safety rules, types of hazards and chemical waste disposal.

Laboratory safety requires a lab coat, goggles, closed shoes, tied-back hair and no eating or drinking. Chemicals should never be tasted, and gases should be smelled by wafting. Physical hazards include broken glass, heat and electricity; chemical hazards include corrosive, toxic, flammable and oxidizing substances; biological hazards include microorganisms.

Chemical waste must be collected in labelled containers according to its nature, not poured randomly into the sink. Acids and bases may need neutralization, and heavy-metal or organic waste requires special disposal. First aid includes washing acid or alkali splashes with plenty of water and reporting accidents immediately.

Important Conceptual Short Questions — Topic 1

1. Why should chemical waste not be poured into the sink?

It may corrode pipes, react with other substances, harm aquatic life or contaminate water supplies.

2. How do physical and chemical hazards differ?

Physical hazards cause harm through energy or objects, such as heat, glass or electricity, while chemical hazards cause harm through toxic, corrosive or reactive properties.

3. Why is wafting safer than directly smelling a gas?

Wafting brings only a small amount of vapour to the nose, reducing exposure to toxic or irritating gases.

4. How should an acid splash on skin be treated first?

Immediately wash the affected area with plenty of running water and inform the teacher or laboratory supervisor.

Topic 2 — Acid-base titration

Q.2 Describe the procedure and precautions in an acid-base titration.

In an acid-base titration, a solution of known concentration is used to find the concentration of another solution. A pipette measures a fixed volume of analyte into a flask, and a burette delivers the titrant. An indicator shows the end point by a colour change.

studysapiens.com

The burette and pipette must be rinsed with the solutions they will contain, air bubbles must be removed, and readings must be taken at eye level. Concordant titres are repeated readings that agree closely. The concentration is calculated using the mole ratio from the balanced equation.

Important Conceptual Short Questions — Topic 2

1. Why is the burette rinsed with the solution it will contain?

Rinsing removes water or previous solution that could dilute or contaminate the titrant and change its concentration.

2. How does an indicator show the end point of a titration?

It changes colour when the pH reaches the range needed to show that neutralization is essentially complete.

3. Why are concordant titres required?

They show that the results are reproducible and reduce random error in the calculated concentration.

4. How can the mole ratio be used to calculate unknown concentration?

Use the balanced equation to relate moles of acid and base, then apply n = cV to find the unknown concentration.

Topic 3 — Tests for anions

Q.3 Describe confirmatory tests for common anions.

Carbonate ions react with dilute acid to release CO_2, which turns limewater milky. Sulfate ions give a white precipitate of $BaSO_4$ with barium chloride in acidic medium. Chloride ions give a white precipitate of AgCl with silver nitrate, soluble in dilute ammonia. Nitrate ions can be confirmed by the brown ring test.

These tests depend on characteristic precipitation or gas-forming reactions. Acidification is important in some tests to remove interfering ions such as carbonate or sulfite.

Important Conceptual Short Questions — Topic 3

1. Why is barium chloride added in acidic medium for sulfate test?

Acid removes carbonate and sulfite ions that could also form white barium precipitates, so the white $BaSO_4$ confirms sulfate.

2. How does limewater confirm carbonate?

CO_2 released from a carbonate reacts with $Ca(OH)_2$ to form insoluble $CaCO_3$, making the limewater milky.

3. Why does AgCl dissolve in ammonia but AgI does not?

AgCl forms a soluble silver-ammine complex with dilute ammonia, while AgI is too insoluble to dissolve under these conditions.

studysapiens.com

4. How does the brown ring test confirm nitrate?

Nitrate is reduced to NO, which forms a brown complex with Fe^{2+} at the junction of the two liquids.

Topic 4 — Tests for cations

Q.4 Describe tests for common basic radicals using sodium hydroxide and ammonia.

Metal cations can be identified by precipitation with NaOH or NH_3 and by behaviour in excess reagent. Al^{3+} and Zn^{2+} give white precipitates soluble in excess NaOH; Zn^{2+} also dissolves in excess ammonia, while Al^{3+} does not. Cu^{2+} gives a blue precipitate that dissolves in excess ammonia to a deep blue solution. Fe^{2+} gives a green precipitate and Fe^{3+} a red-brown precipitate.

Ammonium ions are identified by warming with NaOH, which releases NH_3 gas that turns moist red litmus blue. Flame tests are also useful: Na^+ gives golden yellow, K^+ lilac and Ca^{2+} brick red.

Important Conceptual Short Questions — Topic 4

1. How can excess ammonia distinguish Al^{3+} from Zn^{2+}?

Both give white precipitates with NaOH, but excess ammonia dissolves $Zn(OH)_2$ to form a soluble complex while $Al(OH)_3$ remains insoluble.

2. Why does Cu^{2+} give a deep blue solution with excess ammonia?

Cu^{2+} forms the soluble tetraamminecopper(II) complex, $[Cu(NH_3)_4]^{2+}$, which is deep blue.

3. How can ammonium ion be identified by warming with NaOH?

NaOH releases NH_3 from NH_4^+. The gas has a pungent smell and turns moist red litmus blue.

4. Why are flame tests useful for identifying metal ions?

Some metal ions emit characteristic colours when heated in a flame because their electrons are excited and then release energy of definite wavelengths.

studysapiens.com